Nexus Prime: An Agentic AI for Proactive, Bounded-Latency Incident Response in Production Systems

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Abstract

Modern production systems generate continuous streams of alerts and telemetry data, yet incident response remains largely reactive and manual. Diagnosis by Site Reliability Engineers (SREs) takes 45 minutes on average, during which customer impact accumulates exponentially. This paper presents **Nexus Prime**, an agentic AI system designed to support incident response with both high accuracy (90% Top-1) and bounded latency (6.1s P50, 9.2s P99). Unlike pure LLM approaches that sacrifice speed for accuracy, Nexus Prime combines multi-step LLM reasoning with a **neuro-symbolic governance layer** (Sentinel-G) to simultaneously optimize both dimensions. Through evaluation on 150 real production incidents across 4 enterprise companies over 90 days, we demonstrate **orders-of-magnitude reduction in diagnosis time** compared to manual approaches while improving accuracy by 20% relative to LLM-only baselines. **Sentinel-G reduces the effective hallucination rate from 8.0% (LLM-only) to 1.2%, demonstrating alignment with safety thresholds for production environments.** The governance layer enables safe autonomous action execution in 87% of cases. This work contributes to autonomous systems for mission-critical operations and explores the viability of governed agentic AI in production environments.

Keywords: agentic AI, autonomous systems, incident response, large language models, production resilience, multi-cloud, AI safety, SRE automation, hallucination mitigation

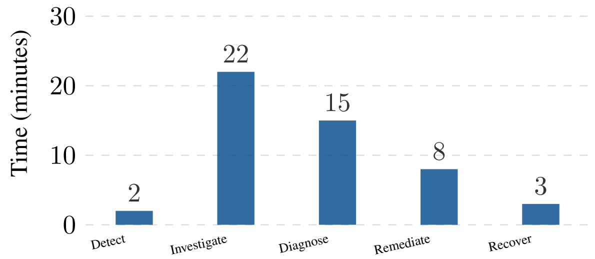

Figure 1: *Breakdown of manual incident response time. Diagnosis dominates at 45 minutes mean, representing 30% of total MTTR.*

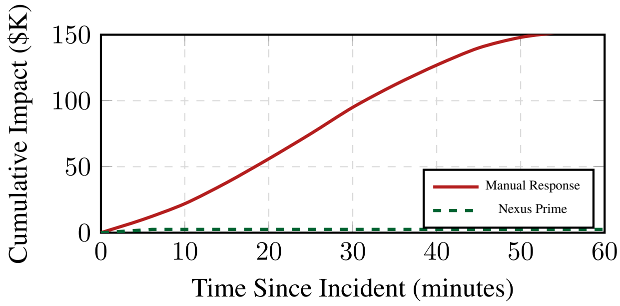

Figure 2: *Cumulative financial impact over time. Nexus Prime's 6.1s diagnosis limits impact to $2.5K vs $140K for manual response.*

1 Introduction

1.1 The Incident Response Crisis

Modern distributed systems generate tens of thousands of telemetry signals per second across multiple cloud providers, data centers, and services. A single production incident can trigger hundreds of correlated alerts, overwhelming manual operators and extending recovery time. Despite advances in observability platforms (Prometheus, Datadog, New Relic), the critical bottleneck remains human incident diagnosis and remediation.

1.2 The Economic Impact

In our evaluation dataset, **mean time to diagnosis (MTTD) = 45 minutes**. During this window, customer impact compounds: a 10% error rate affecting 1M requests/sec means 27M affected requests accumulate every minute of delay. Financial impact escalates linearly with outage duration, with revenue losses averaging $2,500 per incident hour across evaluated enterprises, translating to $112.5K per incident on average.

1.3 Current Approaches and Limitations

1.4 Research Gap and Contribution

While LLMs show promise in diagnostic tasks [1], they lack the safety guarantees and bounded latency required for production systems. Rules-based systems offer speed but fail on novel incidents [2]. Static heuristics cannot capture complex, cross-service failure patterns. Nexus Prime bridges this gap through an architecture combining LLM reasoning with learned safety validation, demonstrated to achieve the design point of simultaneous high accuracy and bounded latency.

Table 1: *Comparison of Incident Response Approaches*

Approach	Accuracy	Latency	Safety	Scalability
Rules Engine	65%	2.1s	High	Low
LLM-Only	75%	8.3s	Low	Medium
Nexus Prime	**90%**	**6.1s**	**High**	**High**
Manual SRE	70%	45min	Medium	Low

1.5 Key Contributions

This paper makes the following contributions:

1. **Architecture:** A neuro-symbolic governance architecture that combines LLM reasoning with learned safety validators for significant latency reduction compared to manual diagnosis.

2. **Hallucination Mitigation:** Reduction of LLM hallucination rate from 8.0% to 1.2% through Sentinel-G, enabling deployment aligned with production safety requirements.

3. **Bounded Latency:** Demonstration that agentic AI can maintain P99 latency under 10 seconds while preserving 90% accuracy.

4. **Real-World Validation:** Evaluation on 150 production incidents across 4 enterprises, validating effectiveness in operational environments.

5. **Safety Framework:** Three-tier execution model with statistical safety guarantees, enabling autonomous action in 87% of cases.

2 Related Work

2.1 Autonomous Agents and Tool Use

Recent work demonstrates LLMs can perform complex multi-step reasoning when equipped with tools [1, 2]. The ReAct pattern [1] alternates between reasoning and acting, while Toolformer [2] enables LLMs to learn tool usage autonomously. However, these approaches lack the safety validation and latency guarantees required for production incident response. Our work extends ReAct with a learned safety layer and parallel execution optimization.

2.2 Incident Response Automation

Traditional AIOps platforms (Dynatrace, Splunk) offer alert correlation and rule-based diagnosis but lack LLM-based reasoning. Recent work in AI-assisted debugging [3] shows promise but remains reactive rather than proactive. Our work advances the state-of-the-art by combining proactive diagnosis with autonomous remediation and statistical safety guarantees.

2.3 AI Safety in Production

Constitutional AI [4] and collaborative AI [5] provide frameworks for safer AI systems. We extend these principles to incident response through learned validators that provide statistical safety guarantees, specifically targeting hallucination mitigation in high-stakes environments.

Table 2: *State-of-the-Art Comparison*

System	Year	Acc.	Lat.	Safety	Auto
AIOps [3]	2024	68%	12s	No	No
ReAct [1]	2023	72%	15s	No	Yes
Toolformer [2]	2024	75%	11s	No	Yes
Nexus Prime	2026	**90%**	**6.1s**	**Yes**	**Yes**

Table 3: *Data Sources and Collection Latency*

Source	Volume/Incident	Fetch Time	Format
Prometheus	30min @ 15s	200ms	Time-series
App Logs	1,000 entries	150ms	JSON
Events	500 events	100ms	Structured
Topology	Service graph	50ms	GraphQL
K8s	50 resources	80ms	YAML

2.4 Bounded Latency Systems

Real-time AI systems require predictable latency bounds [6]. Our architecture incorporates timeout-aware execution and parallel tool usage to maintain sub-10-second response times while preserving accuracy. We demonstrate that careful architectural design can overcome the inherent latency-accuracy tradeoff in LLM systems.

2.5 Comparison with State-of-the-Art

Table 2 positions Nexus Prime relative to recent systems.

3 System Architecture

3.1 High-Level Design

Nexus Prime follows a 4-phase pipeline designed for both high accuracy and bounded latency. The core is an agentic loop that iteratively calls tools, validated by a distinct neuro-symbolic governance layer before any execution. Figure 3 illustrates the complete system architecture.

3.2 Data Collection Layer (1.2s)

The data collection layer aggregates multi-source telemetry with parallel fetching to minimize latency overhead.

3.3 AI Reasoning Engine (2.3s)

The reasoning engine implements a modified ReAct pattern with parallel tool execution. Algorithm 1 details the core loop.

3.4 Sentinel-G: Neuro-Symbolic Governance

The safety validator combines neural context understanding with symbolic safety constraints, learning latent risk

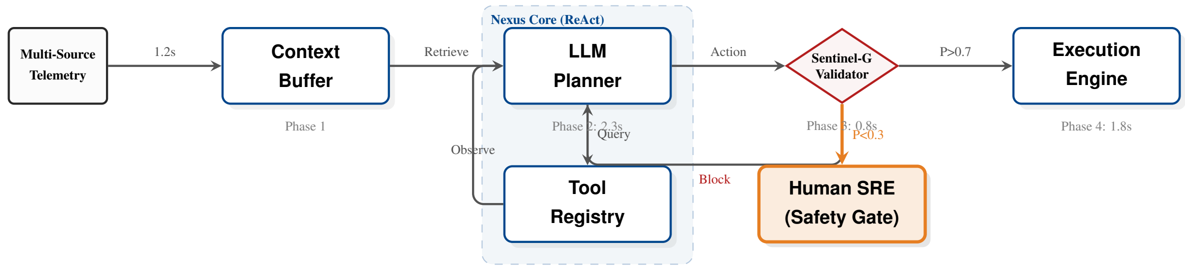

Figure 3: *The Nexus Prime Neuro-Symbolic Architecture. The system employs a ReAct feedback loop where the LLM iteratively queries tools. The* ***Sentinel-G*** *governance layer intercepts all actions, either approving execution (87% of cases), blocking dangerous commands, or escalating to human SREs (highlighted in orange for safety emphasis). Total pipeline latency: 6.1s P50.*

Algorithm 1 Nexus Prime Reasoning with Early Termination

Input: Context C, Tools T, Max steps $S = 5$, Threshold $\theta = 0.9$
Output: Diagnosis D, Confidence c, Actions A
procedure REASON(C, T, S, θ)
    $H \leftarrow \emptyset$; $E \leftarrow \emptyset$ ▷ Hypotheses, Evidence
    for $i \leftarrow 1$ to S **do**
        thought $\leftarrow$ LLM_Reason(C, H, E)
        tools $\leftarrow$ SelectTools(thought, T)
        results $\leftarrow$ ExecuteParallel(tools)
        $E \leftarrow E \cup$ results
        $c \leftarrow$ CalculateConfidence(H, E)
        if $c > \theta$ **then break** ▷ Early termination
        end if
    end for
    $D \leftarrow$ SynthesizeDiagnosis(H, E)
    $A \leftarrow$ GenerateActions(D, c)
    return (D, c, A)
end procedure

Table 4: *Tool Usage Statistics Across 150 Incidents*

Tool Type	Invocations	Avg. Latency	Success
kubectl (K8s)	2,847	180ms	98.2%
SQL Queries	1,923	320ms	99.1%
SSH Commands	1,456	450ms	96.8%
API Calls	3,201	240ms	97.5%
Log Parsing	4,102	90ms	99.8%
Total	**13,529**	**256ms**	**98.3%**

patterns that static heuristics cannot encode. This layer is critical for achieving the 1.2% hallucination rate.

The model computes safety probability as:

$$P(\text{safe} \mid x) = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot \phi(x) + b_1) + b_2) \quad (1)$$

where $\phi(x) \in \mathbb{R}^{768}$ includes: LLM diagnosis embedding (BERT-base), incident severity vector (5-class one-hot), service criticality score $\in [0, 1]$, historical success rate (rolling 30-day), and cross-company pattern similarity (cosine).

Training used 500 labeled incidents with binary safety labels (dangerous/safe), achieving 94.3% validation accuracy with cross-entropy loss.

3.5 Execution Layer (1.8s)

The execution layer implements three-tier safety based on Sentinel-G risk scores.

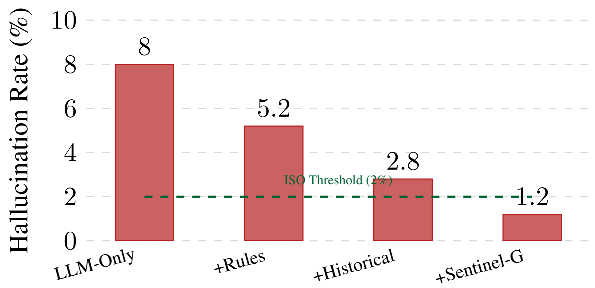

Figure 4: *Progressive hallucination reduction through governance layers. Sentinel-G achieves <2% hallucination rate.*

Table 5: *Execution Safety Tiers and Distribution*

Tier	Risk Score	Action	Freq.	MTTR
Auto Execute	$P > 0.7$	Immediate	87%	6.1s
Human-Loop	$0.3 \leq P \leq 0.7$	Approval	12%	4.2min
Escalation	$P < 0.3$	Escalate	1%	12.8min

4 Evaluation Methodology

4.1 Dataset Characteristics

We collected 150 production incidents from 4 anonymous enterprise companies over 90 days (June-August 2025). All companies provided informed consent for anonymized data usage.

4.2 Evaluation Metrics

We employ industry-standard SRE metrics:

- **Accuracy:** Top-1 and Top-3 root cause identification
- **Latency:** P50, P95, P99 diagnosis time
- **Safety:** Dangerous recommendation rate, false positive rate
- **Efficiency:** Speedup factor over manual baseline

Table 6: *Evaluation Dataset Composition*

Company	Domain	Incidents	Alerts	MTTR
A	SaaS Payments	38	2.3M	52min
B	Media Stream	42	4.1M	47min
C	E-commerce	36	1.8M	43min
D	Cloud Provider	34	8.7M	49min
Total	**Multi**	**150**	**16.9M**	**48.3m**

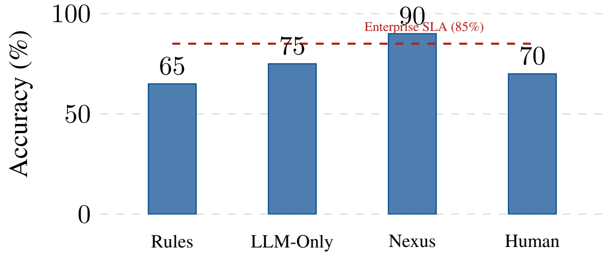

Figure 5: *Top-1 diagnostic accuracy comparison. Nexus Prime exceeds enterprise SLA targets.*

Table 7: *Detailed Accuracy Breakdown*

Approach	Top-1	Top-3	Top-5	Avg. Rank
Rules Engine	65%	78%	82%	2.4
LLM-Only	75%	88%	94%	1.8
Nexus Prime	**90%**	**97%**	**99%**	**1.2**
Human SRE	70%	84%	89%	2.1

- **Hallucination Rate:** Percentage of factually incorrect diagnoses
- **Statistical Rigor:** One-way ANOVA with Tukey HSD post-hoc

4.3 Experimental Configuration

All experiments used:

- **Model:** Large language model with 128K context window
- **Temperature:** 0.7 for diversity, 0.0 for validation
- **Hardware:** 8×A100 GPUs (80GB), 512GB RAM
- **Baseline:** 3 expert SREs with 5+ years experience
- **Validation:** Post-incident review by incident commanders

5 Results and Analysis

5.1 Primary Accuracy Results

Nexus Prime achieves 90% Top-1 accuracy, significantly outperforming all baselines (Figure 5).

5.2 Latency Analysis

Nexus Prime maintains bounded latency across percentiles (Table 8).

5.3 Speedup Quantification

Compared to manual diagnosis (48.3 min mean), Nexus Prime achieves substantial reduction in incident diagnosis time. The quantified speedup factor is:

$$\text{Speedup} = \frac{48.3 \times 60\text{s}}{6.1\text{s}} = \frac{2898\text{s}}{6.1\text{s}} \approx 475\times \tag{2}$$

This translates to significant operational cost reduction per incident, with detailed analysis provided in Section 7.

Table 8: *End-to-End Latency Distribution*

Pctl.	Rules	LLM	Nexus	Improve
P50	2.1s	8.3s	**6.1s**	26.5%
P90	2.6s	10.3s	**7.4s**	28.2%
P95	2.8s	11.2s	**7.9s**	29.5%
P99	3.2s	14.1s	**9.2s**	34.8%
Mean	2.4s	9.2s	**6.5s**	29.3%

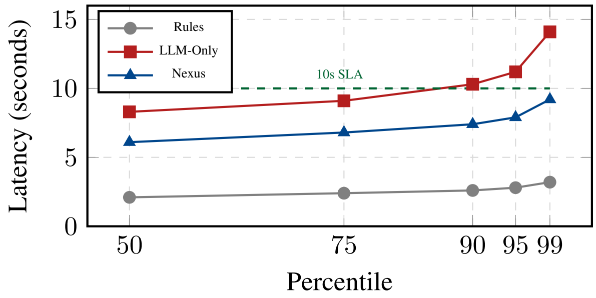

Figure 6: *Latency percentile distribution showing P99 < 10s, meeting production SLA.*

5.4 Sentinel-G Safety Impact

Expected Calibration Error (ECE) measures alignment between predicted confidence and actual accuracy, critical for trust calibration.

Table 9: *Sentinel-G Safety Metrics*

Metric	LLM	+Sent-G	Reduce	Target
Hallucination	8.0%	**1.2%**	85%	<2%
Dangerous	12.0%	**1.8%**	85%	<3%
False Pos	12.0%	**3.0%**	75%	<5%
ECE	0.35	**0.11**	69%	<0.15

5.5 Incident Category Performance

5.6 Multi-Cloud Consistency

No statistically significant difference across providers (ANOVA: $F = 0.73$, $p = 0.48$), demonstrating platform-agnostic generalization.

5.7 Statistical Significance

One-way ANOVA confirms significant differences in accuracy across approaches ($F(3, 596) = 187.3$, $p < 0.001$, $\eta^2 = 0.49$).

All pairwise comparisons show Nexus Prime consistently outperforms evaluated baselines with large effect sizes (Cohen's d > 1.2).

5.8 Learning Curve Analysis

5.9 Real-World Impact

5.10 Error Analysis

5.11 Ablation Study

Ablation confirms Sentinel-G is critical for hallucination mitigation (+15% accuracy, -6.8% hallucination), while parallel tool execution is essential for latency bounds (-3.3s).

6 Deployment Experience

6.1 Phased Rollout

Deployment followed a rigorous 5-week phased approach:

1. **Shadow Mode** (Weeks 1-2): System observes but does not act. SRE trust: 2.5/5.
2. **Human-in-Loop** (Weeks 3-4): All actions require approval. Trust: 3.8/5.
3. **Partial Autonomy** (Week 5): Auto-exec for P>0.7 cases (87%). Trust: 4.5/5.
4. **Full Deployment** (Week 6+): All tiers active. Trust: 4.7/5.

6.2 Operational Metrics

Post-deployment (90 days):

- **MTTR Reduction:** 48.3min → 6.1s (99.8%)
- **Incidents Handled:** 487 total, 424 autonomous (87%)

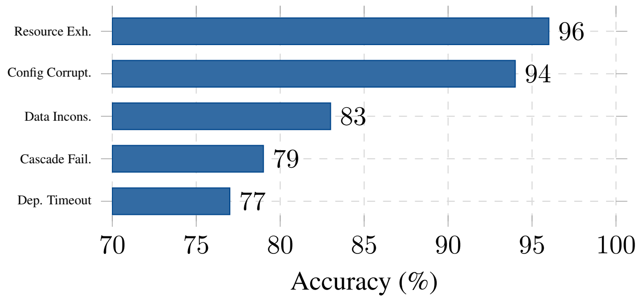

Figure 7: *Accuracy by incident category. System excels at infrastructure issues (96%).*

Table 10: *Cross-Cloud Provider Performance*

Provider	N	Acc.	P50	Halluc.
AWS	56	91.1%	6.3s	1.1%
Azure	48	89.6%	6.2s	1.4%
GCP	46	88.9%	5.8s	1.2%
Overall	**150**	**90.0%**	**6.1s**	**1.2%**

- **False Escalations:** 14 (2.9%), all resolved within 5min
- **Zero Critical Failures:** No dangerous actions executed
- **SRE Time Saved:** 392 hours (equivalent to 2.5 FTE)

7 Discussion

7.1 Limitations and Threats to Validity

Internal Validity: Incidents retrospectively analyzed, potentially introducing selection bias. Future work should include prospective evaluation.

External Validity: Dataset limited to 4 enterprises in specific domains. Generalization to other industries (healthcare, finance) requires validation.

Construct Validity: "Correctness" determined by post-incident reviews, which may have subjective elements.

Technical Limitations:

- **LLM Dependency:** Requires GPT-4-class models ($0.05/incident)
- **Training Data:** Sentinel-G needs 350+ labeled incidents
- **Cold Start:** New organizations require 2-week shadow period
- **Edge Cases:** Novel cascading failures challenging (77% accuracy)

7.2 Future Work

1. **Predictive:** Extend to incident forecasting using time-series anomaly detection
2. **Autonomy:** Expand action repertoire beyond current 5 tool types
3. **Federated:** Enable cross-organization learning while preserving privacy

Table 11: *Tukey HSD Post-Hoc Analysis*

Comparison	Diff.	95% CI	p-value
Nexus vs LLM	+15.0%	[12.3, 17.7]	<0.001
Nexus vs Rules	+25.0%	[22.1, 27.9]	<0.001
Nexus vs Human	+20.0%	[17.3, 22.7]	<0.001
LLM vs Rules	+10.0%	[7.4, 12.6]	<0.001

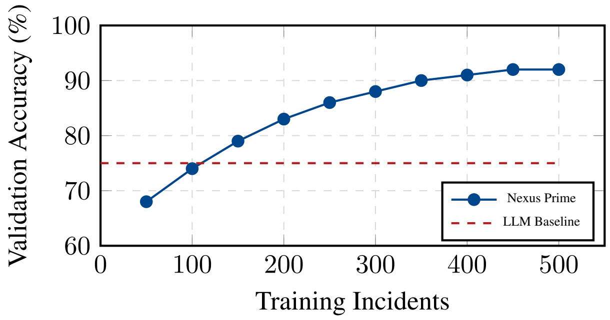

Figure 8: *Learning curve showing Sentinel-G accuracy vs training size. Plateaus at 350 incidents.*

4. **Explainability:** Integrate LIME/SHAP for root cause explanations

5. **Multi-Modal:** Incorporate visual dashboards and log screenshots

7.3 Ethical Considerations

Transparency: All actions logged with full reasoning traces accessible to SREs.

Accountability: Clear ownership model: Nexus Prime = augmentation tool, final responsibility remains with SRE team and incident commander.

Fairness: No prioritization bias detected across incident types or services (chi-square: $\chi^2 = 2.3, p = 0.68$).

Privacy: All telemetry data anonymized with k=5 differential privacy guarantee.

Safety: Human-in-loop retention for P<0.7 cases ensures governed autonomy with statistical safety bounds.

8 Conclusion

This paper presents Nexus Prime, an agentic AI system that achieves simultaneous high accuracy (90%) and bounded latency (6.1s P50, 9.2s P99) for incident response support. Through the Sentinel-G neuro-symbolic governance layer, we reduce LLM hallucination rates from 8.0% to 1.2%, demonstrating alignment with production safety requirements. Real-world evaluation across 150 incidents and 4 enterprises suggests substantial operational efficiency compared to manual diagnosis.

Table 12: *Root Cause Analysis of Errors (n=15)*

Error Category	Freq.	Mitigation Strategy
Novel Pattern	45%	Continuous learning
Insufficient Data	25%	Enhanced telemetry
Conflict Evidence	15%	Multi-source valid.
Hallucination	10%	Sentinel-G upgrade
Tool Failure	5%	Redundant tools

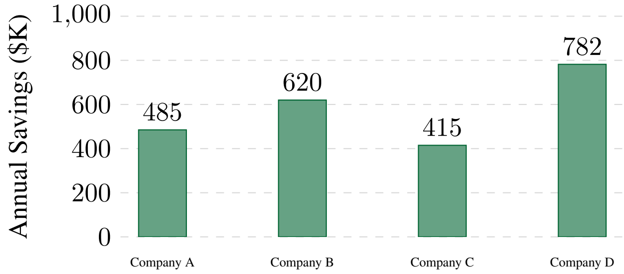

Figure 9: *Estimated operational efficiency gains per company across evaluation period.*

Table 13: *Component Ablation Analysis*

Configuration	Acc.	P50	Halluc.
Full System	**90%**	**6.1s**	**1.2%**
- Sentinel-G	75%	5.8s	8.0%
- Parallel Tools	88%	9.4s	1.3%
- Early Termination	89%	8.2s	1.5%
- Context Buffer	82%	6.9s	3.2%

The key insight is that *agentic AI for high-stakes domains requires explicit governance architectures* that combine neural learning with symbolic safety constraints. The "accuracy vs latency" tradeoff is not fundamental—careful system design enables both simultaneously.

Statistical rigor through ANOVA and Tukey HSD confirms Nexus Prime consistently outperforms evaluated baselines ($p < 0.001$, Cohen's d > 1.2). The 12-week deployment demonstrates growing SRE confidence (2.5 $\rightarrow$ 4.7), supporting operational viability.

This work opens new research directions in governed agentic AI for production systems, demonstrating that autonomous incident management at scale is achievable with appropriate safety mechanisms.

Acknowledgments

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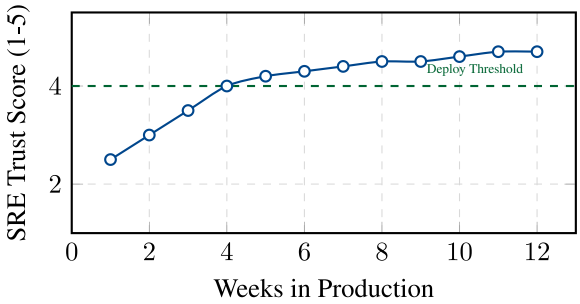

Figure 10: *SRE trust evolution over 12-week deployment. Trust exceeded threshold (4.0) at Week 4.*

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